AI-Face: A Million-Scale Demographically Annotated AI-Generated Face Dataset and Fairness Benchmark*

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Abstract

State-of-the-art AI-generated face detectors exhibit significant performance disparities across demographic groups, particularly in skin tone, gender, and age. Existing datasets either lack comprehensive demographic annotations or cover only limited generative methods, hindering the development and evaluation of fair detectors. To address this challenge, we introduce ***AI-Face****, a million-scale dataset with 1,646,545 face images (400,885 real; 1,245,660 fake) spanning 37 generation methods across deepfake videos, GANs, and diffusion models. AI-Face provides comprehensive demographic annotations for skin tone (using the Monk Skin Tone scale), gender, and age through novel CLIP-based annotators and colorimetric skin-tone estimation. We establish a fairness-first benchmark that evaluates 12 detectors across group and intersectional bias metrics, reliability under post-processing, and cross-domain generalization. Our experiments reveal systematic bias patterns: female and light-skin subgroups exhibit higher false positive rates, and diffusion-model subsets induce the largest fairness gaps. While 10/12 detectors achieve AUC >98%, fairness performance varies significantly, with the best methods achieving skin tone F_{OAE} of 0.88% and intersectional F_{EO} of 18.35%. Cross-domain evaluation shows substantial performance drops (up to -30.9 pts AUC), highlighting generalization challenges. AI-Face establishes a rigorous foundation for developing equitable and robust AI-generated face detectors.*

1. Introduction

The rapid advancement of generative AI has made it increasingly easy to create convincing fake face images and videos. From deepfake videos generated using face-swapping techniques [1, 2] to photorealistic synthetic faces produced by GANs [3] and diffusion models [4], the quality and accessibility of AI-generated content pose significant challenges for media authenticity and digital forensics.

While numerous detection methods have been proposed to combat this threat [5–7], recent studies have revealed a critical limitation: these detectors exhibit systematic biases across demographic groups [8, 9]. Such algorithmic bias not only undermines the reliability of detection systems but also raises serious concerns about fairness and equity, particularly when these systems are deployed in high-stakes applications such as content moderation, criminal investigations, and media verification.

The problem stems from several fundamental challenges in current research. First, existing datasets either lack comprehensive demographic annotations or cover only limited generation methods. For instance, while datasets like A-FF++ [8] provide gender and age labels, they focus exclusively on deepfake videos and lack skin tone annotations. Second, evaluation protocols primarily emphasize utility metrics (accuracy, AUC) while giving insufficient attention to fairness considerations across demographic subgroups. Third, the rapid evolution of generative models has outpaced the development of comprehensive benchmarks that can systematically evaluate detector performance across diverse generation techniques.

To address these limitations, we introduce **AI-Face**, a million-scale demographically annotated dataset that comprehensively covers AI-generated faces from 37 different generation methods across three major categories: deepfake videos, GANs, and diffusion models. Our key contributions are:

- **Comprehensive Dataset**: AI-Face contains 1,646,545 face images with detailed demographic annotations (skin tone via Monk scale, gender, age) across 37 generation methods—the most diverse coverage to date.
- **Novel Annotation Framework**: We develop CLIP-based demographic annotators with fairness-aware training and a colorimetric skin-tone estimator, achieving superior annotation quality validated through extensive user studies.
- **Fairness-First Benchmark**: We establish the first comprehensive fairness benchmark for AI-generated face detection, measuring group bias, intersectional bias, reliability under post-processing, and cross-domain generalization.
- **Systematic Bias Analysis**: Our evaluation of 12 state-of-the-art detectors reveals concerning bias patterns and provides practical guidelines for developing fair and robust

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detection systems.

Our findings reveal that while most detectors achieve high utility (AUC >98%), significant fairness gaps persist across demographic groups. Female and light-skin subgroups consistently exhibit higher false positive rates, and detectors trained on traditional deepfakes struggle with newer diffusion-generated content. These insights highlight the urgent need for fairness-aware detector design and comprehensive evaluation protocols that our benchmark enables.

2. Related Work and Motivation

2.1. AI-Generated Face Detection

The field of AI-generated face detection has evolved rapidly in response to advances in generative technologies. Early detection methods focused on artifact-based approaches, identifying compression artifacts [**?**] and temporal inconsistencies in videos [**?**]. With the emergence of high-quality generators like StyleGAN [3] and face-swapping techniques in FaceForensics++ [1], researchers developed more sophisticated detection approaches.

Current detection methods can be categorized into four main families: (1) **Naive approaches** using standard CNN architectures like Xception [6], EfficientNet [5], and Vision Transformers [7]; (2) **Frequency-domain methods** like F3Net [10] and SPSL [11] that exploit spectral artifacts; (3) **Spatial methods** including SRM [12], UCF [13], and UnivFD [14] that focus on spatial inconsistencies; and (4) **Fairness-enhanced methods** like DAW-FDD [15], DAG-FDD [15], and PG-FDD [16] that explicitly address bias concerns.

2.2. Existing Datasets and Their Limitations

Several datasets have been proposed for AI-generated face detection, each with specific strengths and limitations. Table 4 provides a comprehensive comparison. FaceForensics++ [1] and DFDC [2] focus primarily on deepfake videos with limited demographic coverage. CelebDF-v2 [17] offers higher quality deepfakes but lacks diversity in generation methods. Recent efforts like A-FF++, A-DFDC, and A-DFD [8] add demographic annotations but are limited to specific generation techniques.

More recent datasets attempt broader coverage: DF-Platter [18] includes both deepfakes and GANs with demographic annotations, while GenData [19] focuses on GAN and diffusion models. However, none achieve comprehensive coverage across all three major generation categories (deepfake videos, GANs, diffusion models) while providing detailed demographic annotations for fairness evaluation.

2.3. Algorithmic Fairness in Face Detection

The issue of algorithmic bias in face-related technologies has gained significant attention following revelations of performance disparities across demographic groups in commercial face recognition systems [20**?**]. In the context of deepfake detection, several studies have identified concerning bias patterns. Trinh et al. [9] found that detectors perform differently across gender and skin tone groups. Xu et al. [8] conducted systematic evaluation across multiple datasets, revealing consistent performance gaps.

However, existing fairness evaluations suffer from several limitations: (1) limited demographic coverage (often only gender or age), (2) focus on group-level metrics without considering intersectional bias, (3) lack of reliability assessment under post-processing, and (4) insufficient evaluation of cross-domain generalization. Our work addresses these gaps through a comprehensive fairness framework.

2.4. Motivation for AI-Face

The limitations of existing approaches motivate our development of AI-Face. Current datasets fail to provide the scale, diversity, and demographic coverage necessary for comprehensive fairness evaluation. Specifically:

Scale and Coverage Gap: No existing dataset combines million-scale size with coverage across deepfake videos, GANs, and diffusion models while providing comprehensive demographic annotations.

Evaluation Gap: Existing benchmarks prioritize utility over fairness, lacking systematic evaluation of group bias, intersectional bias, reliability, and cross-domain generalization.

Annotation Quality Gap: Current demographic annotations often suffer from low quality, particularly for challenging cases involving occlusion, lighting variations, or low resolution.

AI-Face addresses these gaps by providing the first million-scale, demographically comprehensive dataset with a fairness-first evaluation framework that enables systematic analysis of bias sources and mitigation strategies.

3. AI-Face Dataset

3.1. Dataset Overview

AI-Face comprises 1,646,545 face images including 400,885 real faces and 1,245,660 AI-generated faces across 28 subsets representing 37 distinct generation methods. The dataset spans three major generation categories: deepfake videos (4 subsets), GANs (10 methods), and diffusion models (8 methods), making it the most comprehensive collection for fairness-aware deepfake detection research. Figure 1 illustrates our data construction pipeline.

3.2. Data Sources and Collection

Real Faces: We aggregate real faces from multiple high-quality sources to ensure diversity and representativeness:

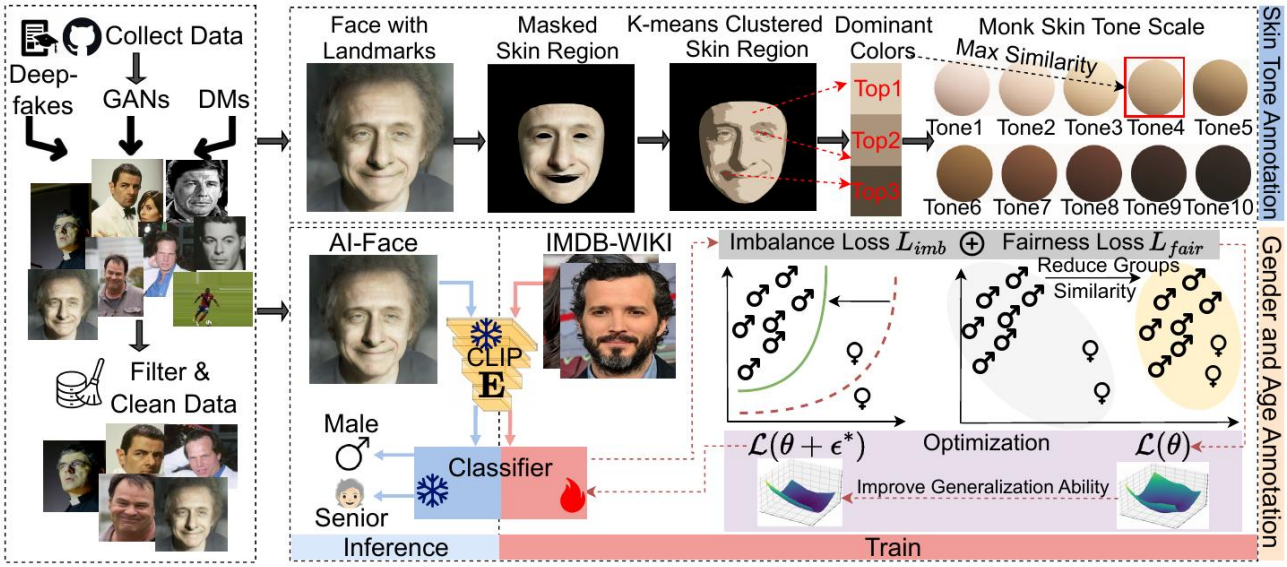

Figure 1. Generation pipeline of our Demographically Annotated AI-Face Dataset. The pipeline includes source data collection, face extraction, demographic annotation using colorimetric skin-tone estimation and CLIP-based gender/age prediction, and quality validation.

FFHQ [3] (70,000 images), IMDB-WIKI [21] (100,000 images), and real partitions from established deepfake datasets including FaceForensics++ [1], DFDC [2], DFD [22], and CelebDF-v2 [17] (230,885 images total).

Deepfake Videos: We include faces extracted from four major deepfake video datasets: FaceForensics++ (5 manipulation methods), DFDC, DFD, and CelebDF-v2, covering diverse face-swapping and reenactment techniques.

GAN-Generated Faces: Our collection includes 10 GAN variants: StyleGAN, StyleGAN2, StyleGAN3, ProGAN, PGGAN, StarGAN, CycleGAN, VQGAN, and others, representing different architectural approaches and training strategies.

Diffusion Model Faces: We incorporate 8 diffusion-based generators: Stable Diffusion v1.5, Stable Diffusion Inpainting, Midjourney, DALL-E 2, DALL-E 3, Imagen, IF, and others, covering both open-source and commercial diffusion models.

All data collection follows strict license compliance, and face detection and cropping are performed using RetinaFace to ensure consistent preprocessing across all sources.

3.3. Demographic Annotation Framework

3.3.1. Skin Tone Estimation

We develop a novel colorimetric approach for skin tone annotation using the Monk Skin Tone (MST) scale [**?**], a 10-point scale designed for inclusive representation. Our pipeline:

1. **Face Landmarking**: Apply FaceMesh to identify facial landmarks and extract skin regions while avoiding hair, eyes, and background areas.

2. **Skin Masking**: Generate precise skin masks using landmark-guided segmentation to focus on representative skin areas.

3. **Color Clustering**: Apply K-means clustering (K=3) on skin pixels in LAB color space to identify dominant skin tones and handle lighting variations.

4. **MST Mapping**: Map dominant colors to the nearest Monk Skin Tone value using perceptual color distance in LAB space.

3.3.2. Gender and Age Prediction

We design CLIP-based demographic annotators that leverage the strong visual representations of CLIP ViT-L/14 [23]. Our approach:

1. **Architecture**: Freeze the pre-trained CLIP ViT-L/14 encoder and add lightweight 3-layer MLP heads for gender (binary) and age (5 categories: 0–18, 19–30, 31–45, 46–60, 60+) prediction.

2. **Training Strategy**: Train on IMDB-WIKI with Vector Scaling loss to handle class imbalance and a novel Sinkhorn-based fairness loss to reduce demographic bias in predictions.

3. **Optimization**: Use Sharpness-Aware Minimization (SAM) to improve generalization across diverse demographic groups and generation methods.

For IMDB-WIKI images, we use provided ground truth labels. For all other images, we apply our trained annotators to generate demographic labels.

Evaluation Type	Dataset	Gender ACC	Gender Precision	Gender Recall	Age ACC	Age Precision	Age Recall
Strategic	A-FF++	8.143	17.583	5.966	37.700	39.459	45.381
	AI-Face	**86.857**	**74.404**	**77.367**	**85.700**	**74.024**	**63.751**
	A-DFDC	21.600	28.604	23.082	33.400	38.011	40.165
	AI-Face	**91.700**	**92.129**	**83.448**	**77.000**	**76.184**	**62.646**
Random	A-Celeb-DF-V2	89.628	90.626	90.494	-		
	AI-Face	**91.206**	**91.474**	**91.767**	-		
	A-DFD	70.900	71.686	74.435	-		
	AI-Face	**92.300**	**91.060**	**91.727**	-		

Table 1. Annotation quality assessment results (%) comparing AI-Face with existing datasets. ACC: Accuracy.

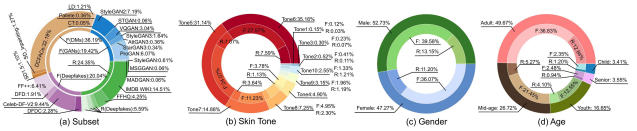

Figure 2. Distribution of face images in the AI-Face dataset across (a) generation subsets, (b) skin tone using Monk scale, (c) gender, and (d) age groups.

3.4. Quality Control and Validation

We conduct extensive quality assessment through human evaluation studies to validate our annotation accuracy:

Study Design: Three annotators independently label stratified samples from AI-Face and existing datasets (A-FF++, A-DFDC, A-DFD, A-CelebDF-v2) using majority vote consensus. We evaluate both "strategic" cases (challenging instances with occlusion, poor lighting, or low resolution) and random samples.

Results: Table 1 shows our annotation quality significantly outperforms existing datasets. For strategic cases, AI-Face achieves gender accuracy of 86.9% vs. 8.1% for A-FF++, and age accuracy of 85.7% vs. 37.7%. For random sampling, our gender accuracy reaches 91.2–92.3% compared to 70.9–89.6% for existing datasets.

3.5. Dataset Statistics and Distribution

Figure 2 presents the demographic distribution across AI-Face. The dataset achieves reasonable balance across demographic groups: skin tone distribution spans the full Monk scale with emphasis on middle tones (4–7), gender distribution is approximately balanced (52% female, 48% male), and age distribution covers all categories with concentration in young adult groups (19–45).

Compared to existing datasets (Table 4), AI-Face provides superior coverage in all dimensions: scale (1.6M+ images), generation diversity (37 methods vs. previous best of 8), and demographic completeness (skin tone, gender, age vs. partial coverage in existing datasets). Figure 3 visualizes this comprehensive advantage across the three key dimensions of demographic annotation, generation category coverage, and number of generation methods.

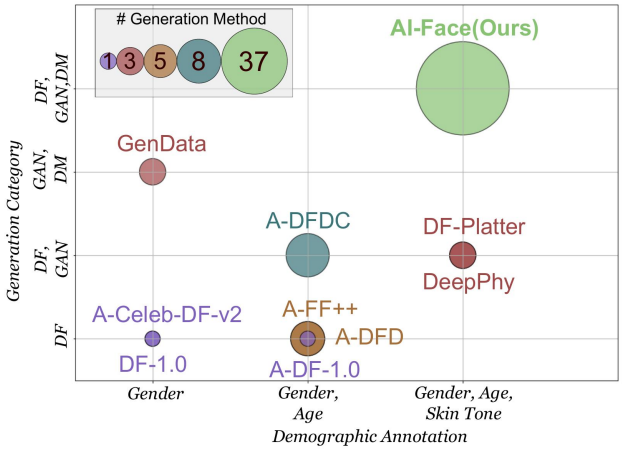

Figure 3. Comparison between AI-Face and other datasets in terms of demographic annotation, generation category coverage, and number of generation methods (circle size).

4. Fairness Benchmark Settings

4.1. Evaluation Framework

Our benchmark establishes the first comprehensive fairness evaluation framework for AI-generated face detection, measuring four key aspects: (1) utility performance, (2) group and intersectional fairness, (3) reliability under post-processing, and (4) cross-domain generalization. This multi-dimensional evaluation provides a holistic view of detector performance beyond traditional accuracy metrics.

4.2. Detector Selection

We evaluate 12 state-of-the-art detectors representing four major families:

Naive Methods: Xception [6], EfficientNet-B4 [5], and ViT-B/16 [7] serve as baseline approaches using standard architectures without domain-specific modifications.

Frequency-Domain Methods: F3Net [10], SPSL [11], and SRM [12] exploit frequency-domain artifacts and spectral inconsistencies common in AI-generated content.

Spatial Methods: UCF [13], UnivFD [14], and CORE [24] focus on spatial artifacts and inconsistencies in AI-generated faces.

Fairness-Enhanced Methods: DAW-FDD [15], DAG-FDD [15], and PG-FDD [16] incorporate explicit fairness constraints during training to mitigate demographic bias.

4.3. Fairness Metrics

We adopt a comprehensive set of fairness metrics covering both group-level and intersectional bias:

Group Fairness Metrics:

- **Demographic Parity** (F_{DP}): Measures differences in positive prediction rates across groups

- **Equalized Odds** (F_{EO}): Evaluates differences in true positive rates across groups
- **Mean Equalized Odds** (F_{MEO}): Averages true positive and false positive rate differences
- **Overall Accuracy Equality** (F_{OAE}): Compares accuracy differences between groups

Individual Fairness: F_{IND} measures consistency of predictions for similar individuals across demographic boundaries.

Intersectional Fairness: We evaluate fairness across intersectional subgroups (e.g., Female-Light, Male-Dark) using the same metrics to capture compound bias effects.

4.4. Utility Metrics

Standard detection performance is measured using:

- **Area Under ROC Curve (AUC)**: Overall classification performance
- **Accuracy (ACC)**: Correct prediction rate at 0.5 threshold
- **Average Precision (AP)**: Area under precision-recall curve
- **Equal Error Rate (EER)**: Error rate when FPR equals FNR
- **False Positive Rate (FPR)**: Rate of incorrectly classified real faces

4.5. Robustness Evaluation

We assess detector reliability under six common post-processing operations that may occur in real-world scenarios:

- **JPEG Compression (JC)**: Quality factor 70
- **Gaussian Blur (GB)**: $\sigma = 1.5$
- **HSV Saturation (HSV)**: $\pm 20\%$ saturation adjustment
- **Brightness Control (BC)**: $\pm 15\%$ brightness adjustment
- **Resizing/Transform (RT)**: 80% scale and restoration
- **Random Crop (RC)**: 85% crop and resize

4.6. Cross-Domain Generalization

To evaluate generalization capability, we test AI-Face trained models on three external datasets:

- **CCv2** [25]: Real faces only (fairness evaluation via demographic parity)
- **DF-Platter** [18]: Mixed deepfake and GAN content
- **GenData** [19]: Focused on GAN and diffusion model content

4.7. Training Protocol

All models use consistent training protocols: 80/20 train/test split, standard data augmentation (random crops, flips, color jittering), Adam optimizer with learning rate 1e-4, batch size 32, and early stopping based on validation AUC. Training is conducted on NVIDIA A100 GPUs with identical hardware and software environments to ensure fair comparison.

Existing Benchmarks	Year	Category			Scope of Benchmark		
		Deepfake Videos	GAN	DM	Utility	Fairness	
						General	Reliability
Loc et al. [9]	2021	✓			✓	✓	
CCv1 [26]	2021	✓	✓		✓	✓	
DeepfakeBench [27]	2023	✓	✓		✓		
CDDB [28]	2023		✓		✓		
Lin et al. [29]	2024	✓	✓		✓		
Le et al. [30]	2024	✓	✓		✓		
DF40 [31]	2024	✓	✓	✓	✓		
Ours	2025	✓	✓	✓	✓	✓	✓

Table 2. Comparison of existing AI-generated face detection benchmarks and ours. Our benchmark is the first to provide comprehensive fairness evaluation including reliability assessment.

5. Experiments and Analysis

5.1. Overall Performance Analysis

Table 3 presents comprehensive results across all evaluated detectors on AI-Face. Our evaluation reveals several key findings:

High Utility Performance: 10 out of 12 detectors achieve AUC $> 98\%$, demonstrating that high detection accuracy is achievable on AI-Face. PG-FDD leads in utility metrics with AUC 99.17%, AP 99.69%, and FPR 10.97%. EfficientNet-B4 shows the best balance with competitive AUC (98.61%) and moderate training time (2h25min/epoch).

Fairness-Utility Trade-offs: Interestingly, fairness-enhanced methods do not uniformly dominate fairness metrics. While PG-FDD achieves the best intersectional F_{EO} (18.35%), UnivFD (spatial method) achieves the best skin tone F_{MEO} (2.58%). This suggests that explicit fairness constraints during training may not always translate to superior fairness performance across all demographic dimensions.

Training Efficiency: Naive methods generally require less training time (1h15min-2h40min/epoch) compared to fairness-enhanced methods (up to 7h20min/epoch for PG-FDD), highlighting the computational cost of fairness-aware training.

5.2. Demographic Bias Analysis

5.2.1. Group-Level Bias Patterns

Our analysis reveals systematic bias patterns across demographic groups:

Skin Tone Bias: Across most detectors, lighter skin tones exhibit lower error rates, with F_{MEO} ranging from 2.58% (UnivFD) to 23.71% (F3Net). PG-FDD achieves the best overall skin tone fairness with F_{OAE} of 0.88%.

Gender Bias: Gender-based disparities are generally smaller than skin tone bias, with F_{MEO} ranging from 3.19% (PG-FDD) to 5.91% (DAG-FDD). Female subgroups consistently show higher false positive rates across most detectors.

Age Bias: Age presents the most significant fairness challenges, with F_{MEO} ranging from 5.44% (UnivFD) to 55.03% (F3Net). Younger age groups typically receive more favorable treatment, while older adults experience higher misclassification rates.

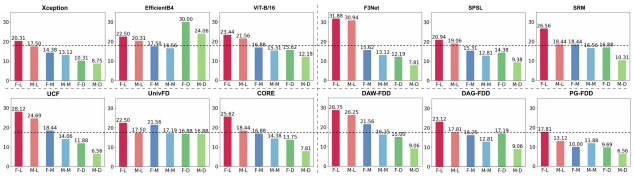

Figure 4. FPR (%) of each intersectional subgroup across different detectors. The dashed line represents the lowest FPR on Female-Light (F-L) subgroup, highlighting performance disparities.

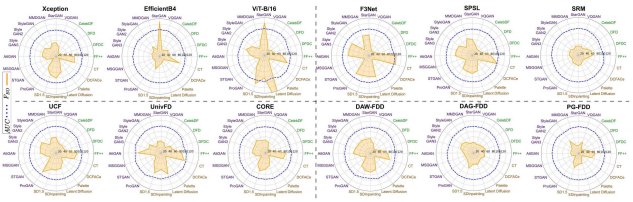

Figure 5. Radar chart visualization of intersectional F_{EO} (%) and AUC (%) of detectors across different generation subsets.

5.2.2. Intersectional Bias

Figure 4 reveals concerning intersectional bias patterns. The Female-Light (F-L) subgroup consistently achieves the lowest FPR across most detectors, serving as a performance ceiling. Male-Dark (M-D) subgroups often experience the highest FPR, with differences exceeding 15 percentage points for some detectors. This intersectional analysis demonstrates that bias effects compound across demographic dimensions.

5.3. Subset-Specific Performance Analysis

Figure 5 presents radar charts showing detector performance across different generation subsets. Key observations:

Diffusion Model Challenge: Diffusion-generated content consistently induces the largest fairness gaps across most detectors. This reflects the relative novelty of diffusion models in training data and their different artifact patterns compared to GANs and deepfakes.

Traditional Deepfake Robustness: Detectors generally perform well on traditional deepfake content (FaceForensics++, DFDC), likely due to extensive representation in training datasets.

Method-Specific Patterns: Different detector families show distinct strength patterns. Frequency-domain methods excel on GAN content but struggle with some diffusion models, while spatial methods show more consistent performance across categories.

5.4. Robustness and Reliability Analysis

Figure 6 shows performance ratios under post-processing. Key findings:

Universal Degradation: All post-processing operations generally degrade both utility and fairness performance, with

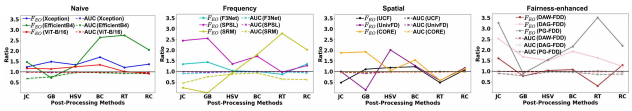

Figure 6. Performance ratio (after vs. before) under different post-processing methods across detector families. Ratios below 1.0 indicate degradation.

JPEG compression and Gaussian blur causing the most significant impacts.

Method-Specific Robustness: Spatial methods show relatively better robustness to geometric transformations, while frequency-domain methods are more sensitive to compression artifacts.

Fairness Volatility: Fairness metrics exhibit higher volatility under post-processing than utility metrics, suggesting that bias mitigation strategies may not transfer well to perturbed inputs.

5.5. Cross-Domain Generalization

Table 5 presents cross-domain evaluation results. Notable findings:

Substantial Performance Drops: AUC drops of up to -30.9 points (EfficientNet-B4 on DF-Platter) demonstrate significant generalization challenges. No detector consistently maintains performance across all external datasets.

Fairness Generalization Gaps: Fairness metrics show even larger degradation than utility metrics during cross-domain transfer. For instance, gender F_{OAE} can increase by over 6 percentage points when transferring from AI-Face to external datasets.

Dataset-Specific Patterns: CCv2 generally allows better utility maintenance but limited fairness evaluation (real faces only). DF-Platter poses the greatest challenge for most detectors, while GenData shows moderate difficulty.

5.6. Training Data Analysis

Figure 7 analyzes the impact of training set size and real/fake ratios on performance:

Scale Benefits: Larger training sets generally improve both utility and fairness, though gains diminish beyond 60% of full data. This suggests AI-Face's scale enables effective detector training.

Optimal Balance: A 1:1 to 1:3 real-to-fake ratio appears optimal for most detectors, balancing representation while avoiding overfitting to either class.

Method Sensitivity: Different detector families show varying sensitivity to these factors, with fairness-enhanced methods generally requiring larger datasets for optimal performance.

Measure	Attribute	Metric	Model Type											
			Naive			Frequency			Spatial			Fairness-enhanced		
			Xception [6]	EfficientB4 [5]	ViT-B/16 [7]	F3Net [10]	SPSL [11]	SRM [12]	UCF [13]	UnivFD [14]	CORE [24]	DAW-FDD [15]	DAG-FDD [15]	PG-FDD [16]
Fairness(%)↓	Skin Tone	F_{MEO}	8.836	8.300	6.264	19.938	8.055	10.002	17.325	**2.577**	10.779	14.118	6.551	6.465
		F_{DP}	9.751	**6.184**	7.728	12.876	9.379	10.897	12.581	8.556	10.317	10.706	8.617	9.746
		F_{OAE}	1.271	4.377	2.168	2.818	1.135	0.915	1.883	2.748	1.332	1.667	1.388	**0.882**
		F_{EO}	12.132	11.062	8.813	23.708	9.789	14.239	21.92	**5.536**	13.069	16.604	7.383	9.115
	Gender	F_{MEO}	3.975	5.385	5.104	4.717	4.411	6.271	5.074	4.503	5.795	5.510	5.910	**3.190**
		F_{DP}	1.691	1.725	1.344	1.864	1.827	1.957	1.736	**1.190**	2.154	2.015	2.151	1.252
		F_{OAE}	**0.975**	1.487	1.803	1.129	1.037	1.772	1.451	1.622	1.389	1.325	1.420	1.071
		F_{EO}	4.143	5.863	6.031	4.870	4.534	6.78	5.510	5.408	5.931	5.696	6.066	**3.702**
	Age	F_{MEO}	27.883	6.796	14.937	38.801	27.614	24.843	47.500	**5.436**	33.882	45.466	15.229	14.804
		F_{DP}	10.905	11.849	11.839	14.906	11.232	11.570	17.049	15.249	12.564	14.106	**9.633**	10.467
		F_{OAE}	7.265	**2.856**	6.838	10.116	7.270	6.524	11.652	3.793	8.760	11.878	5.533	5.009
		F_{EO}	42.216	**10.300**	30.795	55.032	40.943	38.528	67.545	14.148	48.729	64.384	30.182	29.585
	Intersection	F_{MEO}	10.505	17.586	9.384	21.369	10.379	15.142	20.134	**6.119**	15.34	16.565	12.178	9.578
		F_{DP}	14.511	**8.607**	11.535	17.175	13.259	15.186	17.03	14.026	14.301	14.088	11.705	14.697
		F_{OAE}	2.536	8.461	4.928	4.870	**2.464**	3.998	3.536	6.287	2.775	3.547	4.035	3.062
		F_{EO}	24.315	25.114	27.443	47.783	21.679	30.112	43.376	20.255	28.84	33.122	26.295	**18.348**
	Individual	F_{IND}	10.338	25.742	**0.022**	1.872	2.518	7.621	0.767	3.523	0.041	3.772	0.901	0.780
Utility(%)	-	AUC↑	98.583	98.611	98.69	98.714	98.747	97.936	98.082	98.192	98.579	97.811	98.771	**99.172**
		ACC↑	96.308	94.203	94.472	95.719	**96.346**	95.092	95.151	93.651	96.224	95.426	95.722	96.174
		AP↑	99.350	99.542	99.571	99.453	99.356	99.172	99.273	99.400	99.360	99.015	99.498	**99.694**
		EER↓	5.149	6.689	6.372	5.256	**4.371**	6.483	7.708	7.633	5.145	7.063	5.499	4.961
		FPR↓	12.961	20.066	16.426	14.679	13.661	15.746	13.646	18.550	13.410	16.670	14.844	**10.971**
Training Time / Epoch			**1h15min**	2h25min	2h40min	1h18min	1h20min	3h10min	5h05min	4h	1h16min	1h25min	1h17min	7h20min

Table 3. *Overall performance comparison of difference methods on the AI-Face dataset. The best performance is shown in* ***bold****.*

Dataset	Year	Face Images		Generation Category			#Generation Methods	Source of Real Images	Demographic Annotation		
		#Real	#Fake	Deepfake Videos	GAN	DM			Skin Tone	Gender	Age
DF-1.0 [32]	2020	2.9M	14.7M	✓			1	Self-Recording		✓	
DeePhy [33]	2022	1K	50.4K	✓	✓		3	YouTube	✓	✓	✓
DF-Platter [18]	2023	392.3K	653.4K	✓	✓		3	YouTube	✓	✓	✓
GenData [19]	2023	-	20K		✓	✓	3	CelebA [34]		✓	
A-FF++ [8]	2024	29.8K	149.1K	✓			5	YouTube		✓	✓
A-DFD [8]	2024	10.8K	89.6K	✓			5	Self-Recording		✓	✓
A-DFDC [8]	2024	54.5K	52.6K	✓	✓		8	Self-Recording		✓	✓
A-Celeb-DF-v2 [8]	2024	26.3K	166.5K	✓			1	Self-Recording		✓	
A-DF-1.0 [8]	2024	870.3K	321.5K	✓			1	Self-Recording		✓	✓
AI-Face	2025	400K	1.2M	✓	✓	✓	37	FFHQ [3], IMDB-WIKI [21], real from FF++ [1], DFDC [2], DFD [22],Celeb-DF-v2 [17]	✓	✓	✓

Table 4. *Quantitative comparison of existing datasets with ours on demographically annotated AI-generated faces.*

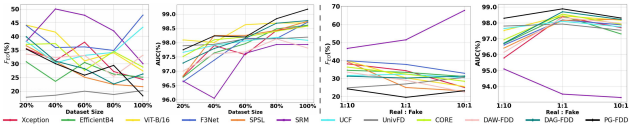

Figure 7. Impact of training set size (**Left**) and ratio of real to fake images (**Right**) on detectors' intersectional F_{EO} (%) and AUC (%).

6. Discussion and Implications

6.1. Key Insights

Our comprehensive evaluation reveals several critical insights for the AI-generated face detection community:

Fairness-Utility Paradox: While achieving high utility (AUC $>$ 98%) is feasible, ensuring fairness across demographic groups remains challenging. Even fairness-enhanced methods show significant bias in certain dimensions, suggesting that current mitigation strategies are insufficient.

Intersectional Complexity: Simple group-level fairness evaluation underestimates bias severity. Intersectional analysis reveals compound discrimination effects that affect minority subgroups disproportionately, with FPR differences exceeding 15 percentage points between best and worst-performing subgroups.

Generation Method Impact: Different generation techniques induce distinct bias patterns. Diffusion models, being relatively new, present particular challenges for fair detection, while traditional deepfakes show more predictable bias patterns.

Robustness Fragility: Post-processing operations not only degrade utility but also destabilize fairness properties, suggesting that bias mitigation strategies may not transfer well to real-world deployment scenarios.

6.2. Practical Recommendations

Based on our findings, we offer several recommendations for developing fair AI-generated face detectors:

Multi-Dimensional Evaluation: Adopt comprehensive evaluation protocols that assess group fairness, intersectional bias, reliability, and cross-domain generalization rather than focusing solely on utility metrics.

Balanced Training Data: Ensure representative demographic coverage across all generation methods, particularly emerging techniques like diffusion models that may introduce novel bias patterns.

Fairness-Aware Architecture Design: Develop detection architectures that explicitly account for demographic

Model Type	Detector	Dataset							
		CCv2 [25]		DF-Platter [18]			GenData [19]		
		Fairness(%)↓	Utility(%)↑	Fairness(%)↓		Utility(%)↑	Fairness(%)↓		Utility(%)↑
		F_{OAE}	ACC	F_{OAE}	F_{EO}	AUC	F_{OAE}	F_{EO}	AUC
Naive	Xception	1.006(+0.031)	86.465(-9.843)	6.836(+5.861)	9.789(+5.646)	81.273(-17.310)	2.539(+1.564)	13.487(+9.344)	96.971(-1.612)
	EfficientB4	4.077(+0.259)	82.980(-11.223)	8.786(+7.299)	12.370(+6.507)	67.694(-30.917)	3.304(+1.817)	**1.995** (-3.686)	93.213(-5.398)
	ViT-B/16	2.167(+0.364)	81.489(-12.983)	**0.015** (-1.788)	12.373(+6.342)	76.050(-22.640)	3.164(+1.361)	9.610(+3.579)	88.253(-10.437)
Frequency	F3Net	5.743(+4.614)	87.867(-7.852)	3.521(+2.392)	**6.445** (+1.575)	85.112(-13.602)	1.188(+0.059)	16.306(+11.436)	91.603(-7.111)
	SPSL	**0.601** (-0.436)	80.006(-16.340)	5.109(+4.072)	7.842(+3.308)	82.175(-16.572)	1.385(+0.348)	9.261(+4.272)	**98.838** (+0.091)
	SRM	7.000(+5.228)	79.768(-15.324)	3.823(+2.051)	6.567(-0.213)	66.401(-31.535)	3.281(+1.509)	7.907(+1.127)	90.049(-7.887)
Spatial	UCF	2.169(+0.718)	**93.009** (-2.142)	8.687(+7.236)	17.068(+11.558)	80.821(-17.261)	3.513(+2.062)	10.529(+5.019)	87.778(-10.304)
	UnivFD	7.625(+6.003)	67.983(-25.668)	4.540(+2.918)	9.950(+4.542)	76.443(-21.749)	1.645(+0.023)	3.848(-1.560)	94.418(-3.774)
	CORE	4.410(+3.021)	83.328(-12.896)	7.741(+6.352)	17.348(+11.417)	77.226(-21.353)	3.759(+2.370)	23.289(+17.358)	98.408(-0.171)
Fairness-enhanced	DAW-FDD	4.726(+3.401)	84.685(-10.741)	5.536(+4.211)	13.667(+7.791)	81.807(-16.004)	1.443(+0.118)	10.228(+4.532)	97.854(+0.043)
	DAG-FDD	2.364(+0.944)	83.918(-11.804)	3.064(+1.644)	22.203(+16.137)	75.206(-23.565)	**0.714** (-0.706)	10.332(+4.266)	92.108(-6.663)
	PG-FDD	1.513(+0.442)	92.852(-3.322)	4.565(+3.494)	9.717(+6.015)	**85.271** (-13.901)	3.063(+1.992)	9.479(+5.777)	93.329(-5.843)

Table 5. *Fairness generalization results based on the gender attribute. The smallest performance changes (in parentheses) and the best performance are in red and in **bold**, respectively. Only F_{OAE} fairness metric and ACC metric are used in CCv2 due to all samples are real.*

factors during feature learning, rather than treating fairness as an afterthought.

Robust Bias Mitigation: Design fairness interventions that maintain effectiveness under post-processing and domain shift, addressing real-world deployment challenges.

6.3. Limitations and Future Work

While AI-Face advances the state of fairness evaluation in AI-generated face detection, several limitations remain:

Demographic Scope: Our current annotation framework focuses on skin tone, gender, and age. Future work should expand to include other relevant demographic factors such as ethnicity, disability status, and cultural background.

Generation Evolution: As new generation techniques emerge, AI-Face will require continuous updates to maintain relevance. Developing adaptive annotation and evaluation frameworks represents an important future direction.

Fairness Definitions: Our metric selection reflects current fairness definitions in computer vision. Alternative fairness concepts from other domains may provide additional insights.

7. Conclusion

This work introduces AI-Face, the first million-scale demographically annotated dataset for fair AI-generated face detection. Through comprehensive evaluation of 12 state-of-the-art detectors across utility, group fairness, intersectional bias, reliability, and cross-domain generalization, we reveal significant performance disparities across demographic groups that existing evaluation protocols fail to capture.

Our key contributions include: (1) AI-Face dataset with 1.6M+ images spanning 37 generation methods with high-quality demographic annotations; (2) novel CLIP-based annotation framework with fairness-aware training; (3) comprehensive fairness benchmark establishing new evaluation standards; and (4) systematic bias analysis providing actionable insights for fair detector development.

The findings highlight urgent challenges in current AI-generated face detection: while high utility is achievable, ensuring fairness across demographic groups remains difficult. Intersectional bias effects are particularly concerning, with performance gaps exceeding 15 percentage points between subgroups. Post-processing and domain shift further destabilize fairness properties, complicating real-world deployment.

AI-Face establishes a rigorous foundation for developing equitable and robust AI-generated face detectors. By providing both the dataset and evaluation framework, we enable the research community to systematically address bias in detection systems and advance toward more inclusive AI technologies. Future work should expand demographic coverage, adapt to emerging generation techniques, and develop bias mitigation strategies that maintain effectiveness across diverse deployment scenarios.

The imperative for fair AI systems has never been greater as AI-generated content becomes increasingly prevalent in society. AI-Face provides the tools and benchmarks necessary to ensure that detection systems serve all demographic groups equitably, contributing to a more trustworthy and inclusive digital future.

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